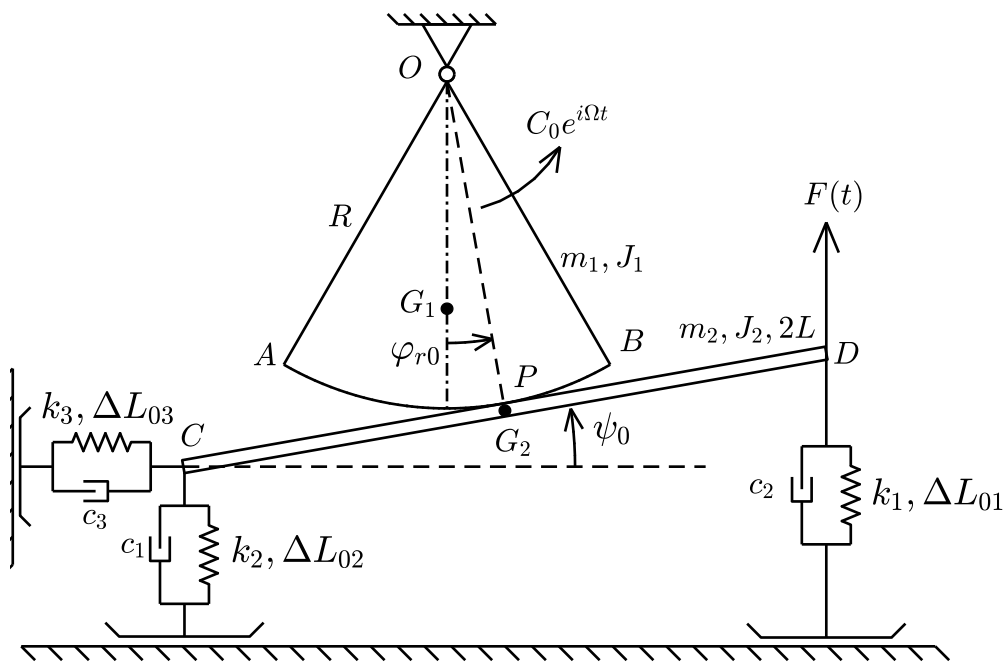


Dynamics of Mechanical Systems
June 17th, 2019
Standard problem (grading up to 30/30)



$m_1 = 5 \text{ kg}$	$\varphi_{r0} = \pi/18$
$m_2 = 5 \text{ kg}$	$F_{01} = 200 \text{ N}$
$J_1 = 0.05 \text{ kgm}^2$	$F_{02} = 150 \text{ N}$
$J_2 = 0.1 \text{ kgm}^2$	$F_{03} = 100 \text{ N}$
$L = 0.25 \text{ m}$	$\varphi_1 = \pi/6$
$r = OG_1 = 0.15 \text{ m}$	$\varphi_2 = 0$
$R = 0.25 \text{ m}$	$\varphi_3 = -\pi/6$
$\Delta L_{01} = -7.2\text{e-}3 \text{ m}$	$\Omega_1 = 4\pi$
$\Delta L_{02} = -7.4\text{e-}3 \text{ m}$	$\Omega_2 = 8\pi$
$\Delta L_{03} = 8.5\text{e-}4 \text{ m}$	$\Omega_3 = 16\pi$
$k_1 = 5\text{e}3 \text{ N/m}$	$C_{02} = 10 \text{ Nm}$
$k_2 = 5\text{e}3 \text{ N/m}$	$\varphi_T = -\pi/3$
$k_3 = 5\text{e}3 \text{ N/m}$	$\Omega_T = 25\pi$
$c_1 = 1 \text{ Ns/m}$	
$c_2 = 1 \text{ Ns/m}$	
$c_3 = 1 \text{ Ns/m}$	

The mechanical system above lays in the vertical plane and is shown in its equilibrium position. The CD bar rolls without sliding on the circular sector ABO. Being P the contact point between bar and circular sector, in the static configuration P lays in the middle of the CD bar and the relative angle φ_{r0} between the axis of the circular sector OG_1 and the line OP is $\pi/18$.

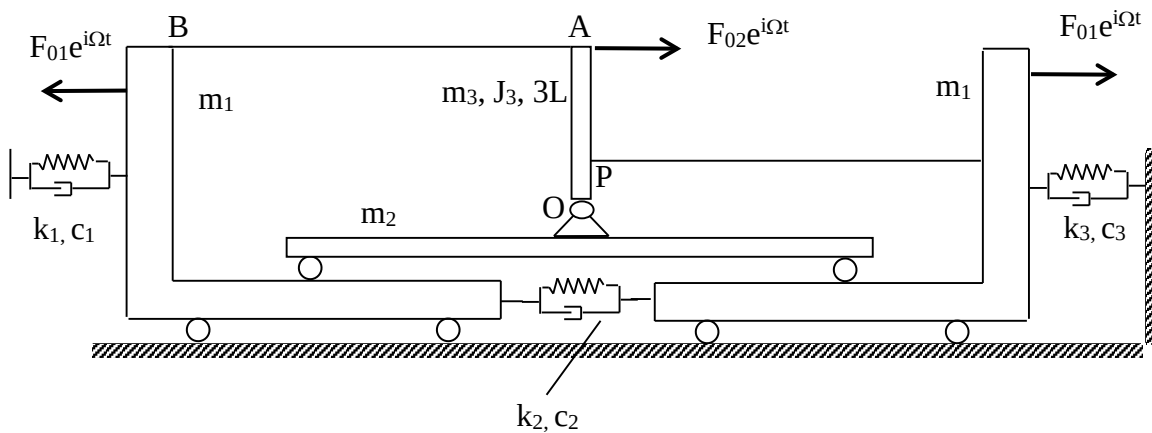
The students are requested to:

1. Write their surname, name, matriculation number and SNxxx code (see note below) on the top of first page of each sheet, leaving blank field "Nr." Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;
2. Write the linearized equations of motion around the given equilibrium configuration;
3. Compute the natural frequencies and modes of vibrations. This answer shall be provided in written on paper;
4. Compute the Frequency Response Function (in the range 0 – 15 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$. Output: the horizontal component of the displacement of point G_1 and the absolute rotation of the CD bar. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 15 Hz, step = 0.01 Hz) applying ad point D the harmonic vertical force $F(t) = F_0 e^{i\Omega t}$. Output: the horizontal component of the constraint force in the hinge O and the elastic and damping force transmitted by the spring and damper system (k_1, c_1). Save the graphical results as SNxxx3.fig and SNxxx4.fig
6. Compute the time history ($T = 1.0 \text{ s}$, time step = 0.001 s) applying the periodic force $F(t) = \sum_{k=1}^3 F_{0k} e^{i(\Omega_k t + \varphi_k)}$ and the harmonic torque $C_{02} e^{i(\Omega_T t + \varphi_T)}$. Output: the horizontal component of the constraint force in the hinge O. Save the graphical result as SNxxx5.fig;

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test June 17th 2019*
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.

Dynamics of Mechanical Systems
June 17th, 2019
Simplified problem (grading up to 22/30)



$m_1 = 10 \text{ kg}$ $k_1 = 1000 \text{ N/m}$ $c_1 = 1 \text{ Ns/m}$
 $m_2 = 5 \text{ kg}$ $k_2 = 1000 \text{ N/m}$ $c_2 = 1 \text{ Ns/m}$
 $m_3 = 3 \text{ kg}$ $k_3 = 1000 \text{ N/m}$ $c_3 = 1 \text{ Ns/m}$
 $J_3 = 0.01 \text{ kgm}^2$
 $L = 0.25 \text{ m}$
 $2AP = PO$

The mechanical system above lays in the vertical plane and is shown in its equilibrium position.

The students are requested to:

1. Write their surname, name, matriculation number and SNxxx code (see note below) on the top of first page of each sheet, leaving blank field "Nr." Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;
2. Write the linearized equations of motion around the given equilibrium configuration;
3. Compute the natural frequencies and modes of vibrations. This answer shall be provided in written on paper;
4. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying the two forces $F_{01}e^{i\Omega t}$ in phase opposition. Output: the displacement of the m_2 mass and the rotation of the OA bar. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying the horizontal force $F_{02}e^{i\Omega t}$ applied at point A. Output: the force transmitted by the spring and damper system (k_2, c_2) and the horizontal component of the displacement of point A. Save the graphical results as SNxxx3.fig and SNxxx4.fig.
6. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying the force $F_{02}e^{i\Omega t}$. Output: the dynamic component of the tensile load in the AB string. Save the graphical results as SNxxx5.fig

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

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7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.